Practical No 05. Adapter Based Views

A) Develop an Android application using ListView with ArrayAdapter and implement item click handling.

Project Name: ListViewDemo

Program Code:

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="10dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Programming Languages"
        android:textSize="24sp"
        android:textStyle="bold"
        android:padding="10dp" />

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

Filename: :- Main Activity.java

```
package com.example.listviewdemo;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
```

```
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private ListView listView;
    private final String[] languages = {
            "Java",
            "Python",
            "C",
            "C++",
            "Kotlin",
            "PHP",
            "JavaScript",
            "Swift"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);

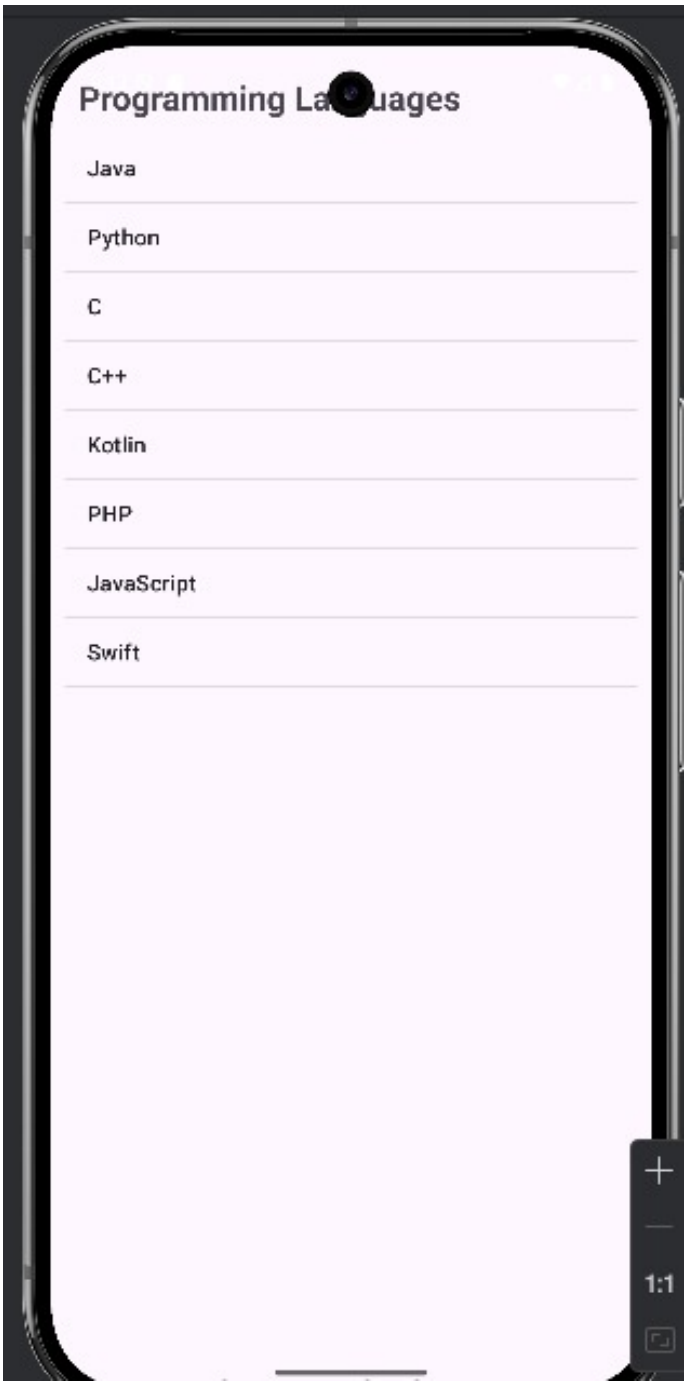

        ArrayAdapter<String> adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                languages
        );

        listView.setAdapter(adapter);

        listView.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> parent, View view, int position, long id) {
                String item = languages[position];
                Toast.makeText(
                        MainActivity.this,
                        "Selected: " + item,
                        Toast.LENGTH_SHORT
                ).show();
            }
        });
```

```
    }
}
```

Output: ListViewDemo



B) Develop an Android application using GridView with Custom Adapter and image selection handling.

Project Name: GridViewDemo

Program Code:

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center_horizontal"
   android:padding="10dp">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:paddingTop="20dp"
      android:paddingBottom="10dp"
      android:text="Image Gallery"
      android:textSize="24sp"
      android:textStyle="bold" />

   <GridView
      android:id="@+id/gridView"
      android:layout_width="match_parent"
      android:layout_height="0dp"
      android:layout_weight="1"
      android:numColumns="2"
      android:verticalSpacing="10dp"
      android:horizontalSpacing="10dp"
      android:stretchMode="columnWidth" />

</LinearLayout>
```

Filename: grid_item.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="10dp">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="120dp"
        android:layout_height="120dp"
        android:scaleType="centerCrop" />

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:paddingTop="5dp" />

</LinearLayout>
```

Filename: CustomAdapter.java

```
package com.example.gridviewdemo;

import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.ImageView;
import android.widget.TextView;

public class CustomAdapter extends BaseAdapter {

    private final Context context;
    private final String[] imageNames;
    private final int[] images;
    private final LayoutInflater inflater;

    public CustomAdapter(Context context, String[] imageNames, int[] images) {
        this.context = context;
        this.imageNames = imageNames;
```

```
    this.images = images;
    this.inflater = LayoutInflater.from(context);
  }

  @Override
  public int getCount() {
    return imageNames.length;
  }

  @Override
  public Object getItem(int position) {
    return imageNames[position];
  }

  @Override
  public long getItemId(int position) {
    return position;
  }

  @Override
  public View getView(int position, View convertView, ViewGroup parent) {
    ViewHolder holder;


    if (convertView == null) {
      convertView = inflater.inflate(R.layout.grid_item, parent, false);

      holder = new ViewHolder();
      holder.imageView = convertView.findViewById(R.id.imageView);
      holder.textView = convertView.findViewById(R.id.textView);

      convertView.setTag(holder);
    } else {
      holder = (ViewHolder) convertView.getTag();
    }

    holder.imageView.setImageResource(images[position]);
    holder.textView.setText(imageNames[position]);

    return convertView;
  }

  private static class ViewHolder {
    ImageView imageView;
```

```
    TextView textView;
  }
}
```

Filename: Main Activity.java

```
package com.example.gridviewdemo;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.GridView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  private GridView gridView;

  private final String[] imageNames = {
      "City",
      "Pikachu",
      "Chiron",
      "Krishna"
  };

  private final int[] images = {
      R.drawable.image1,
      R.drawable.image2,
      R.drawable.image3,
      R.drawable.image4
  };

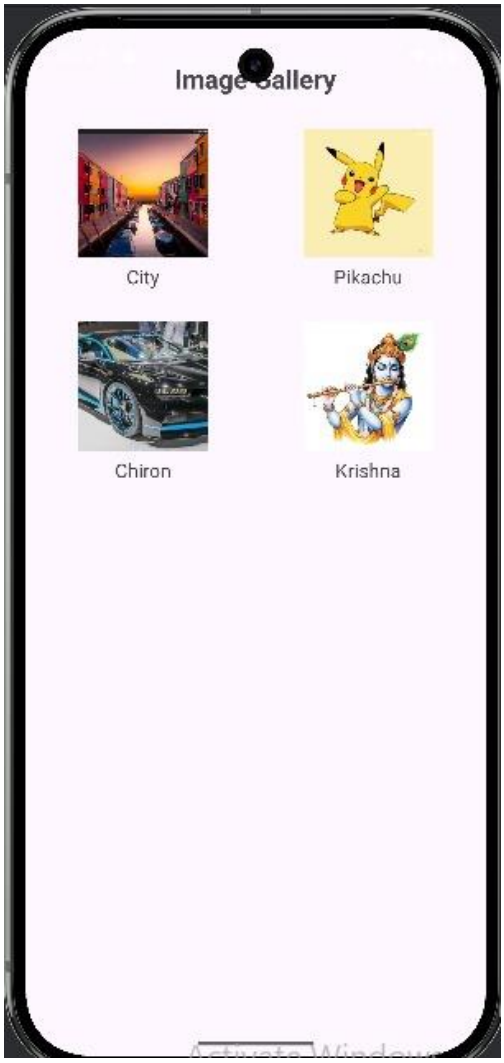
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    gridView = findViewById(R.id.gridView);

    CustomAdapter adapter = new CustomAdapter(this, imageNames, images);
    gridView.setAdapter(adapter);
```

```
    gridView.setOnItemClickListener(new AdapterView.OnItemClickListener() {
      @Override
      public void onItemClick(AdapterView<?> parent, View view, int position, long id) {
        Toast.makeText(
            MainActivity.this,
            "Selected: " + imageNames[position],
            Toast.LENGTH_SHORT
        ).show();
      }
    });
  }
}
```

Output: GridViewDemo



C) Develop an Android application implementing RecyclerView as a modern alternative to ListView.

Project Name: RecyclerViewDemo

Program Code:

Filename: Main Activity.java

```
package com.example.recyclerviewdemo;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

public class MainActivity extends AppCompatActivity {

  private RecyclerView recyclerView;

  private final String[] languages = {
      "Java",
      "Python",
      "C",
      "C++",
      "Kotlin",
      "PHP",
      "C#",
      "Swift",
      "JavaScript",
      "Go",
      "Ruby"
  };

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    recyclerView = findViewById(R.id.recycleView);
    recyclerView.setLayoutManager(new LinearLayoutManager(this));
```

```
        MyAdapter adapter = new MyAdapter(languages);
        recyclerView.setAdapter(adapter);
    }
}
```

Output: RecyclerViewDemo



